1.

(1) 不妨 $\lambda_1 n \in \mathbb{Z}$, $\lambda_1 > H(X \mid Y) + \varepsilon$, λ_2 同理, $\lambda_1 + \lambda_2 > H(X,Y) + \varepsilon$. E_1 是任意一个将 X^n 均匀映到 $\{0,1\}^{\lambda_1 n}$ 的函数, E_2 同理. 令
$M = \{(x,y) \in (XY)^n : \left| -\frac{\log P(x,y)}{n} - H(X,Y) \right| < \varepsilon\}$,

$$D(\hat{x},\hat{y}) := \begin{cases} (x,y), & \exists!(x,y),\ E_1(x) = \hat{x} \wedge E_2(y) = \hat{y} \wedge (x,y) \in M; \\ \perp, & \text{otherwise.} \end{cases}$$

则

$$\begin{aligned} \Pr_{x,y}[D(\hat{x},\hat{y}) \neq (x,y)] &\leq \Pr_{x,y}[(x,y) \notin M] + \Pr_{x,y}[\exists x'\ \exists y'\ E_1(x') = \hat{x} \wedge E_2(y') = \hat{y} \mid (x,y) \in M] \\ &< \mathrm{e}^{-\mathcal{O}(n\varepsilon^2)} + |M| 2^{-(\lambda_1+\lambda_2)n} \\ &= \mathrm{e}^{-\mathcal{O}(n\varepsilon^2)} + 2^{-n(\lambda_1+\lambda_2-H(X,Y)-\varepsilon)} \\ &= 2^{-\Theta(n)}. \end{aligned}$$

(2) (a) 若 $\lambda_1 < H(X \mid Y)$, 不妨 Y 上编码 $D(E_2(y)) = y$ 充分强, 则

$$\begin{aligned} \lambda_1 &< \frac{H(D(E_1(X^n), E_2(Y^n)) \mid Y^n)}{n} \\ &= \frac{H(D_1(E_1(X^n), Y^n) \mid Y^n)}{n} \\ &\leq \frac{H(E_1(X) \mid Y)}{n} \\ &\leq H(X \mid Y). \end{aligned}$$

随 n 增大, 错误率显然不能被控制. (b) 对称地成立. (c) 不妨编码 $(E_1, E_2) = E(X^n, Y^n)$ 充分强, 后续同理.

2.

(1) 弱于 (2).

(2) 设 $\ell > 2d$, E 在 $\{0,1\}^n \to \{0,1\}^\ell$ 均匀随机地选取. 对均匀随机选取的 $c_1, c_2 \in \{0,1\}^\ell$, 有

$$\Pr[\Delta(c_1,c_2) < d] = \Pr\left[\frac{\Delta(c_1,c_2)}{\ell} < \frac{d}{\ell}\right] \overset{\text{Chernoff}}{\leq} \exp\left(-\ell \cdot d\left(\frac{d}{\ell} \parallel \frac{1}{2}\right)\right).$$

任意取定 $x \in \{0,1\}^n$, $c := E(x)$, 有

$$\begin{aligned} \Pr_E\left[\min_{x_1 \neq x_2} \Delta(E(x_1), E(x_2)) < d\right] &\leq \Pr_E\left[\sum_{x' \neq x} \Delta(c, E(x')) < d\right] \\ &\leq 2^n \exp\left(-\ell \cdot d\left(\frac{d}{\ell} \parallel \frac{1}{2}\right)\right) \\ &= 2^{n-\ell(1-h(d/\ell))}. \end{aligned}$$

希望

$$\begin{aligned} \ell - \ell \cdot h(d/\ell) > n &\Leftrightarrow \ell > n + \ell \cdot h(d/\ell) \\ &\Leftarrow \ell > n + \frac{\ln \frac{d}{\ell}}{\ln 2} \cdot \ell \ln\left(1 - \frac{d}{\ell}\right). \end{aligned}$$

而 $\lim_{\ell\to+\infty}\ell\ln\left(1-\frac{d}{\ell}\right)$ 存在, 所以 $\alpha=-\ln 2\cdot\inf_{\ell>2d}\left(\ell\ln\left(1-\frac{d}{\ell}\right)\right)>0$ 存在, 此时

$$\ell>n+\alpha d\ln\frac{\ell}{d}\Rightarrow\ell-\ell\cdot h(d/\ell)>n\Rightarrow\Pr_E\left[\min_{x_1\neq x_2}\Delta(E(x_1),E(x_2))<d\right]<0,$$

即存在 $[\ell,n,d]$-纠错码.

3.

(1) Reed-Solomon 编码.

(2) 只需证 $\ell=n+d-2$ 时不存在. 由于 $|\mathbb{Z}_p^\ell|/|\mathbb{Z}_p^n|=p^{d-2}$, 鸽巢原理给出存在一个 $x\in\mathbb{Z}_p^n$, 使得 $|\{c:D(c)=x\}|\le p^{d-2}$. 这意味对 $c_0=E(x)$, 存在 $\Delta(c_0,c)=d-1$ 使得 $D(c)\neq x$, 与纠错码性质矛盾.

4.

(1) 考虑每个 $S_r=\left\{x\in\Omega:\log\frac{Q(x)}{P(x)}=r\right\}$, 我们希望

$$\begin{cases}\Pr_{X\sim P}[\mathcal{D}(X)\to 1\mid X\in S_r]=\Pr_{X\sim P}[\mathcal{D}'(r)\to 1\mid X\in S_r],\\ \Pr_{X\sim Q}[\mathcal{D}(X)\to 0\mid X\in S_r]=\Pr_{X\sim Q}[\mathcal{D}'(r)\to 0\mid X\in S_r].\end{cases}$$

对于左侧, 由于 $Q(x)/P(x)$ 在 S_r 上相等, P 和 Q 在 condition on $X\in S_r$ 时将 scale up 到相同分布. 因此

$$\Pr_{X\sim P}[\mathcal{D}(X)\to 1\mid X\in S_r]=1-\Pr_{X\sim Q}[\mathcal{D}(X)\to 0\mid X\in S_r]=:f_r.$$

只需取 $\mathcal{D}'(r)\to\mathrm{Bern}(f_r)$ 即可.

(2) 设 $x,y\in\Omega$ 满足 $Q(x)/P(x)<Q(y)/P(y)$, 若 $\mathcal{D}(x)\to\mathrm{Bern}(p)$, $\mathcal{D}(y)\to\mathrm{Bern}(q)$, 则这两点对 $\varepsilon_{\mathrm{FP}}$ 的贡献为

$$\begin{gathered}P(x)p+P(y)q\\ Q(x)(1-p)+Q(y)(1-q)\\ Q(x)+Q(y)+x\end{gathered}$$

[UPD 26/4/1] 这次 skip 了, 不会喵.